Tertiary reaction: The secondary products either decompose or react to give final products

Decomposition of the secondary:

$$\underset{\text{Nitrous acid}}{3\,HNO_2} \longrightarrow \underset{\text{Nitric acid}}{HNO_3} + 2\,\underset{\text{Nitric oxide}}{NO} + H_2O$$

$$\underset{\text{Nitrous acid}}{2\,HNO_2} \longrightarrow \underset{\text{Dinitrogentrioxide}}{N_2O_3} + H_2O$$

$$\underset{\text{Hypo nitrous acid}}{H_2N_2O_2} \longrightarrow \underset{\text{Nitrous oxide}}{N_2O} + H_2O$$

Reaction of secondary products:

$$HNO_2 + NH_3 \longrightarrow N_2 + 2H_2O$$

$$HNO_2 + NH_2OH \longrightarrow N_2O + 2H_2O$$

$$HNO_2 + HNO_3 \longrightarrow 2NO_2 + H_2O$$

Examples:

Copper reacts with nitric acid in the following manner

$$3Cu + 6HNO_3 \longrightarrow 3Cu(NO_3)_2 + 6(H)$$

$$6(H) + 3HNO_3 \longrightarrow 3HNO_2 + 3H_2O$$

$$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$$

overall reation

$$3Cu + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

The concentrated acid has a tendency to form nitrogen dioxide

$$Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

Magnesium reacts with nitric acid in the following way

$$4Mg + 8HNO_3 \longrightarrow 4Mg(NO_3)_2 + 8[H]$$

$$HNO_3 + 8(H) \longrightarrow NH_3 + 3H_2O$$

$$HNO_3 + NH_3 \longrightarrow NH_4NO_3$$

overall reaction

$$4Mg + 10HNO_3 \longrightarrow 4Mg(NO_3)_2 + NH_4NO_3 + 3H_2O$$

If the acid is diluted we get N_2O

$$4Mg + 10HNO_3 \longrightarrow 4Mg(NO_3)_2 + N_2O + 5H_2O$$

Uses of nitric acid:

3. Nitric acid is used as a oxidising agent and in the preparation of aquaregia.
4. Salts of nitric acid are used in photography ($AgNO_3$) and gunpowder for firearms. ($NaNO_3$)

Evaluate yourself :

Write the products formed in the reaction of nitric acid (both dilute and concentrated) with zinc.

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